

TOPOLOGY THAT LEARNS:

Investigating Mycelial and Physarum Networks as a Conceptual Blueprint for Streaming Time-Series Prediction

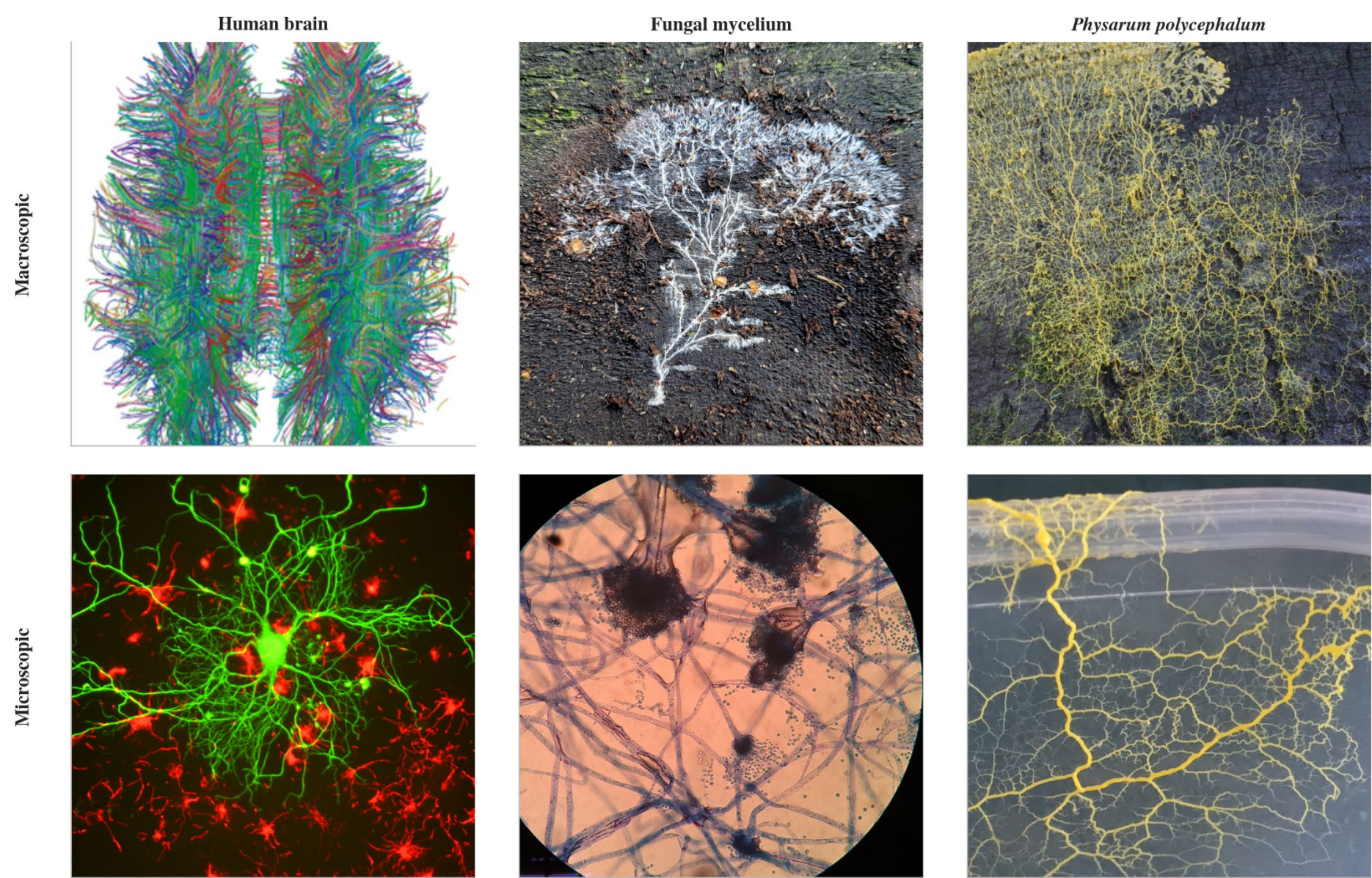
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INTRODUCTION

Streaming time-series models must learn from continuously changing data, retain useful past regimes, and operate within limited computation. However, many forecasting systems rely on largely fixed architectures, making them vulnerable to conceptual drift. Inspired by fungal mycelium and *Physarum polycephalum*, we propose an online transition graph that stores recurring dynamical patterns in its structure. Using hourly temperature data, we test whether this self-organizing topology can recover seasonal dynamics and improve prediction.

BIOLOGICAL INSPIRATION



Most computational models draw inspiration from the human brain, but it is not the only biological network capable of adaptation. Fungal mycelium and *Physarum polycephalum* form decentralized transport networks that continuously reorganize through local activity. Frequently used pathways strengthen, inactive pathways weaken, and dormant pathways can reactivate. In these systems, memory is embodied in network structure.

The Tero conductivity model formalizes this process:

$$\frac{dD_{ij}}{dt} = f(|Q_{ij}|) - \gamma D_{ij}$$

where D_{ij} is pathway conductivity, Q_{ij} is flow, and γ controls decay.

For online streaming prediction, these biological principles could provide a way to couple **learning** and **structure**.

RESEARCH QUESTION

Can local reinforcement and decay cause an online transition network to self-organize into a topology that reflects the recurring structure of a changing dynamical system?

We examine three emergent properties:

- **Structural persistence**
- **Selective memory**
- **Emergent organization**

MODEL OVERVIEW

Biological flow network	Streaming prediction model
Region or resource location	Dynamical state
Transport pathway	Temporal transition
Flow through a pathway	Repeated aligned traversal
Conductivity	Transition strength
Pathway decay	Reduced influence of inactive states
Dormant pathway	Stored but currently inactive regime

The model is an **online transition graph** that re-imagines the Tero equation as an edge property. Conductivity is a weight rather than the capacity of a biological tube.

1. Represent the current dynamical state

Causally smooth the signal with the Matérn-5/2 Kalman model, then form a delay-coordinate embedding:

$$\mathbf{e}_t = [\dot{x}_t, \dot{x}_{t-L}, \dot{x}_{t-2L}]^T$$

2. Store recurring trajectories as graph structure

Each new embedded state becomes part of a directed temporal graph. Transitions retain traversal history, expected signal increment, and conductivity. Recent trajectories are compared with existing nodes; similar states are merged.

3. Adapt memory through reinforcement and decay

Frequently re-traversed, well-aligned transitions strengthen; inactive transitions decay and lose influence.

$$D_{uv} \leftarrow (1 - \gamma)D_{uv} + \alpha Q_t \quad \text{for traversed } (u, v)$$

4. Retrieve matching history and forecast

Match the current trajectory to graph states, weight outgoing historical increments by alignment and conductivity, and add the expected increment to the current filtered level:

$$\hat{y}_{t+1} = \hat{x}_t + \frac{\sum_v Q_v \text{inc}(v)}{\sum_v Q_v}$$

EVALUATION

Hourly 2-meter air temperature from Sacramento, California, was streamed sequentially through the model. **2020–2022** was used to build and adapt the graph online, **2023**: held out for one-step-ahead evaluation.

The model was compared with four local baselines:

- seasonal naïve, persistence, drift, Kalman one-step

We use **MAE**, **RMSE**, **MAPE**, and examination of internal model structure for evaluation.

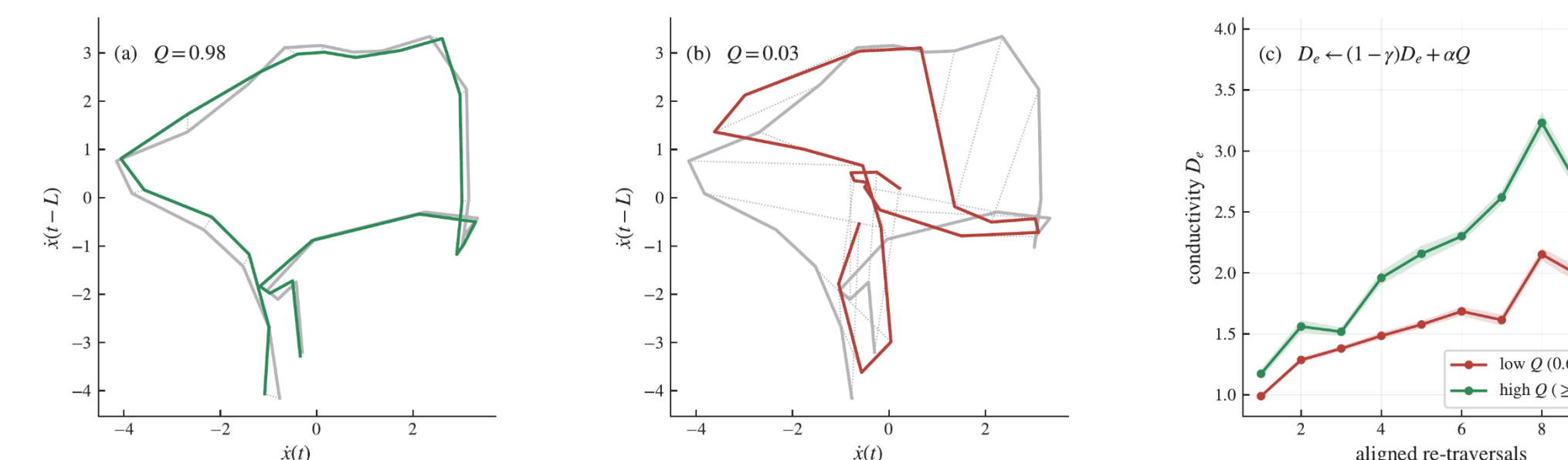
RESULTS AND NETWORK ANALYSIS

Performance Compared to Baselines

Method	RMSE	MAE	MAPE
Seasonal naïve	2.331	1.804	30.5%
Persistence	1.518	1.134	9.1%
Drift	0.830	0.567	5.3%
Kalman	0.815	0.562	5.1%
Model	0.638	0.412	4.2%

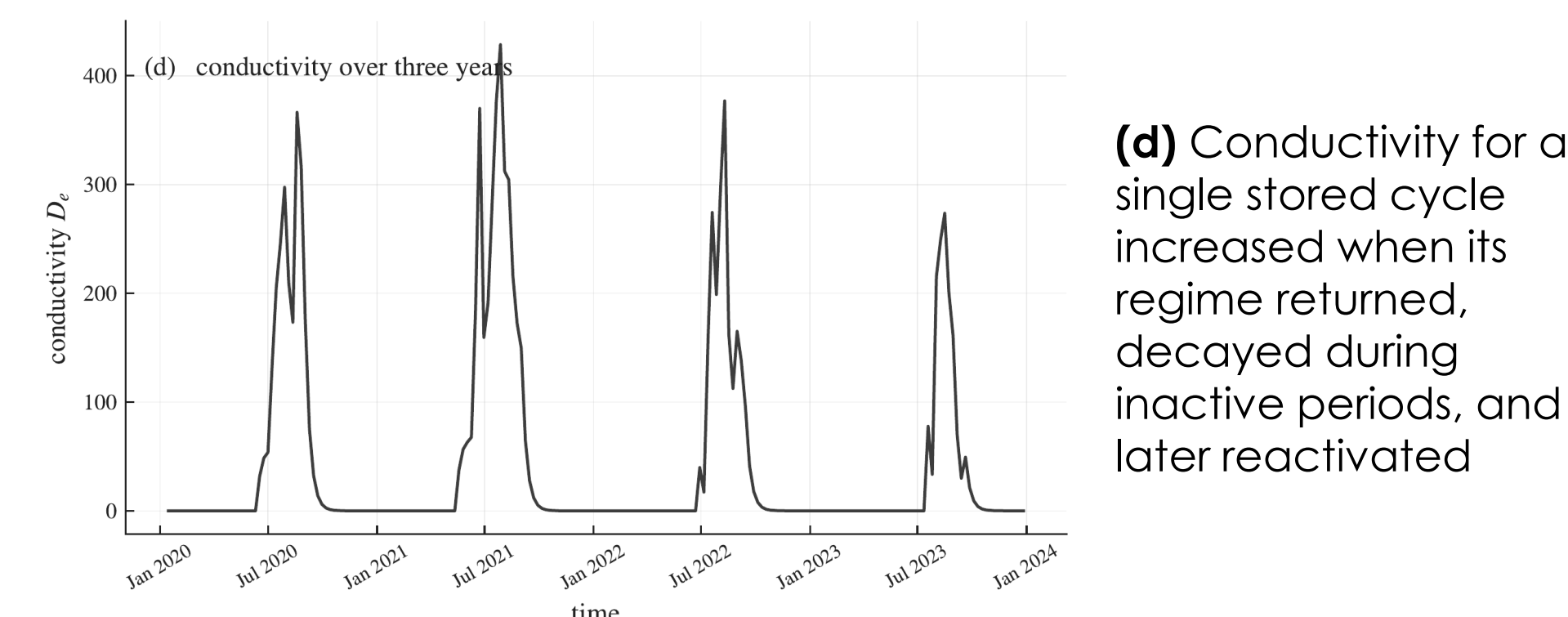
The proposed model achieved the best performance across all metrics. Compared with the one-step Kalman baseline, it reduced RMSE by **21.7%**, MAE by **26.7%**, and MAPE by **17.6%**.

Alignment Scoring

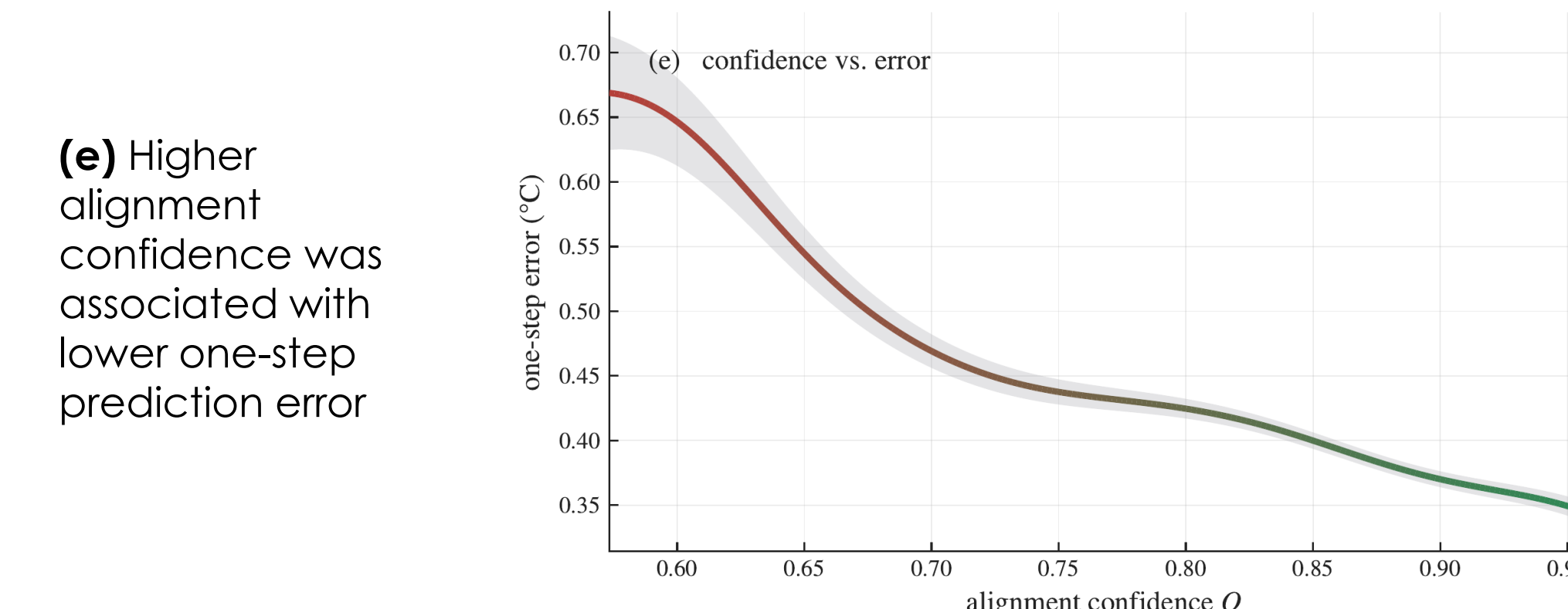


A well-aligned trajectory $Q = 0.98$ (a), poorly aligned trajectory $Q = 0.03$ (b), better-aligned re-traversals accumulate greater conductivity (c)

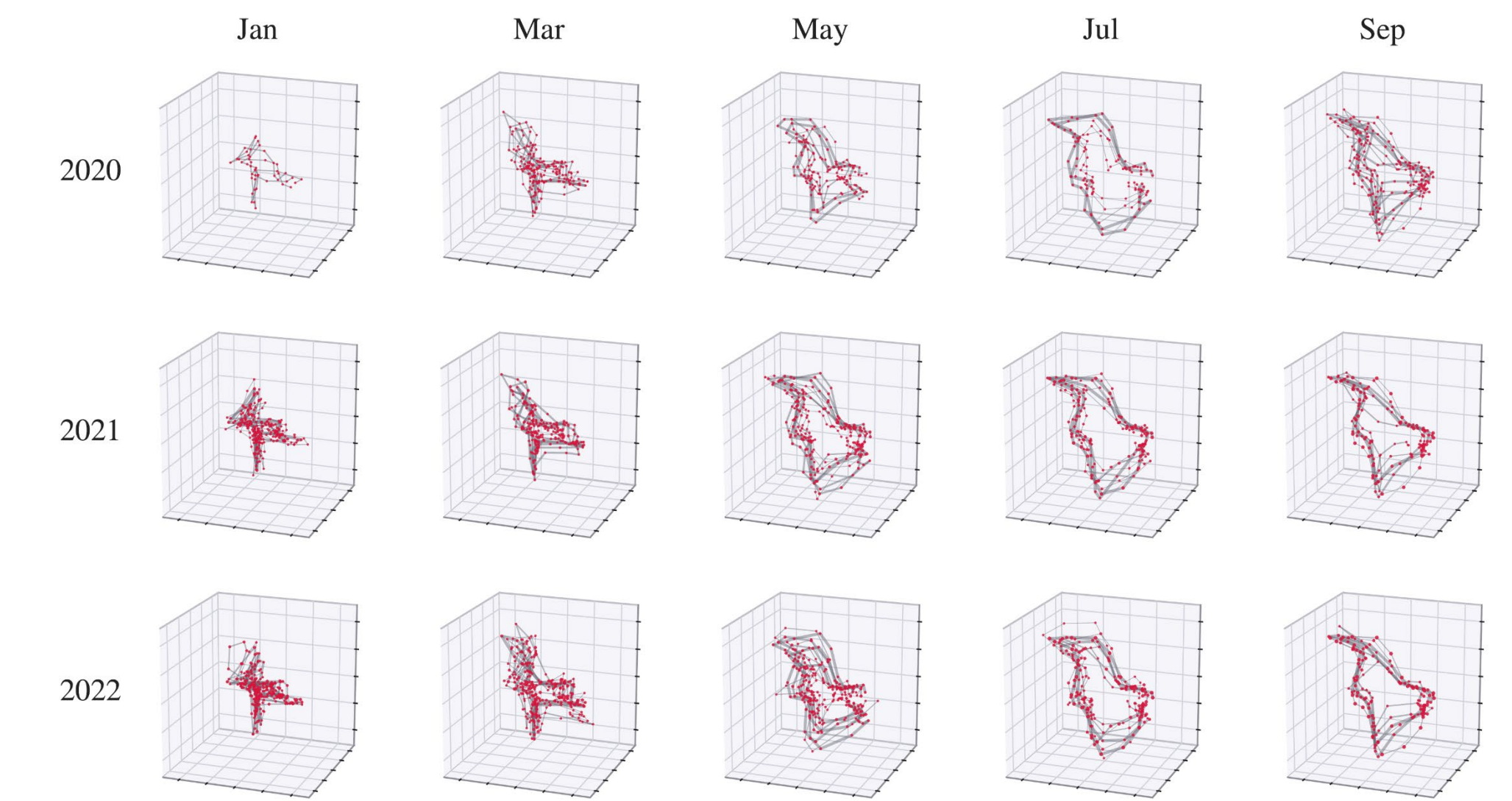
Dormancy and Reactivation



Alignment Confidence vs. Error

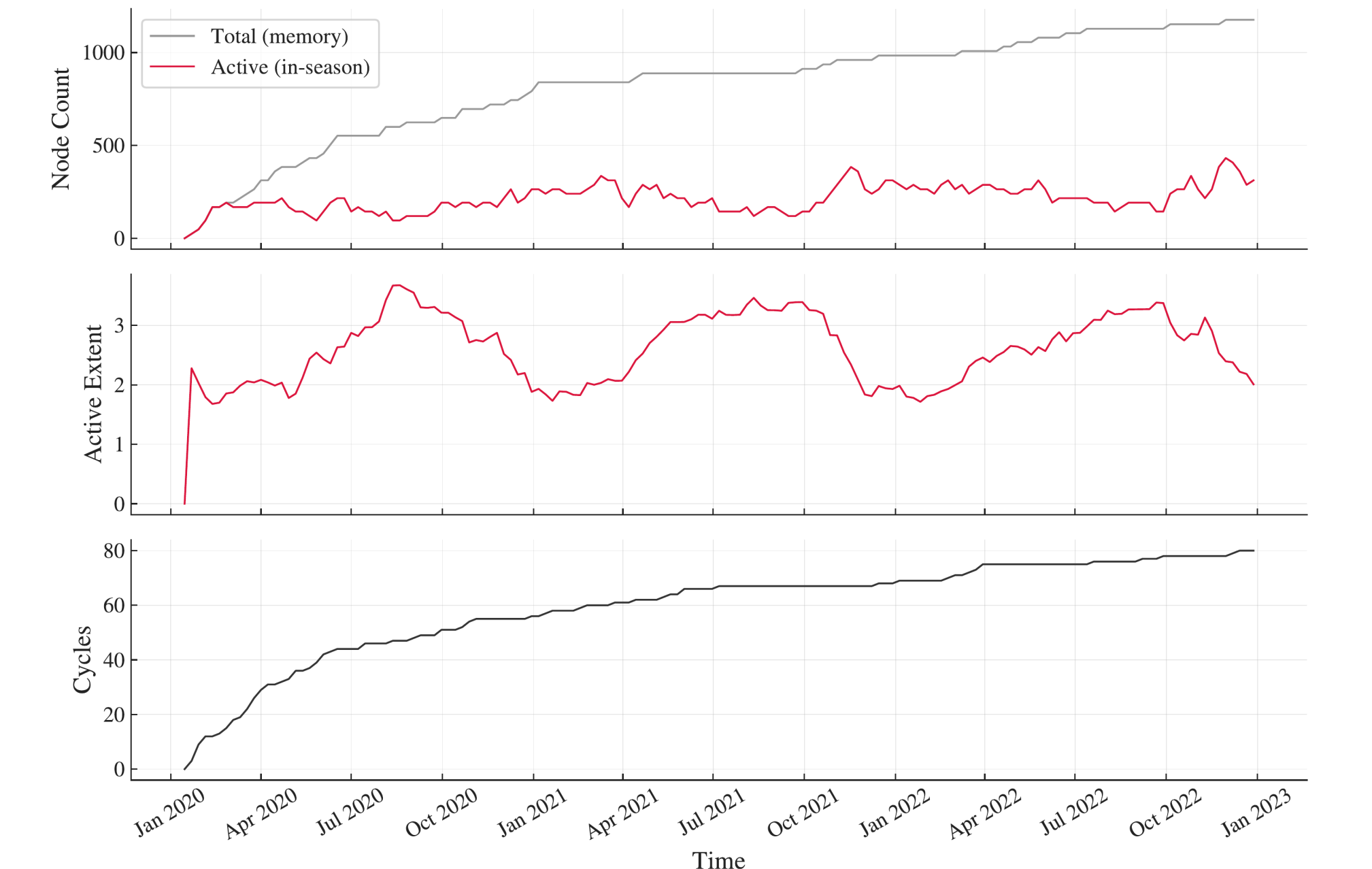


Recurrence of Structural Memory



The model's active memory changes shape over time.

Graph Topology Over Time



The graph grew to about **1,180 nodes** and **80 seasonal cycles**. The active network was about **61% larger in summer**.

DISCUSSION

In addition to improving forecast accuracy, the results show that the model formed a meaningful separation between **persistent memory** and **current activity**. The graph could retain previously learned regimes without allowing all stored structure to influence every prediction. The seasonal reactivation of dormant pathways suggests relevance can be reduced temporarily and restored when similar dynamics return. The results also indicate that the topology carries information about the model's own certainty. When the current trajectory closely matched stored structure, predictions were more reliable.

CONCLUSIONS

This study provides initial evidence that unconventional computing principles can provide practical mechanisms for adaptive forecasting. Extending this approach to multivariate streams, constrained hardware, richer biological rules, and formal theoretical analysis could help establish topology-adaptive computing as a broader framework for real-time learning in non-stationary environments.